17. Design and Implement a Watermarking Technique to Insert a Watermark into the Original Image Using OpenCV

CODE:

```
import cv2
import matplotlib.pyplot as plt

# Read the original image
img = cv2.imread('image.jpg')

if img is None:
    print("Error: Original image not found!")
else:
    # Create a copy of the original image
    watermarked = img.copy()

    # Define watermark text
    watermark = "COPYRIGHT"

    # Set font properties
    font = cv2.FONT_HERSHEY_SIMPLEX
    position = (30, img.shape[0] - 30)
    font_scale = 1
    color = (255, 255, 255)
    thickness = 2

    # Insert watermark into the image
    cv2.putText(
        watermarked,
        watermark,
        position,
        font,
        font_scale,
        color,
        thickness,
        cv2.LINE_AA
    )

    # Save the watermarked image
    cv2.imwrite('watermarked_image.jpg', watermarked)

    # Display original and watermarked images
    plt.figure(figsize=(10, 5))

    plt.subplot(1, 2, 1)
    plt.imshow(cv2.cvtColor(img, cv2.COLOR_BGR2RGB))
    plt.title('Original Image')
    plt.axis('off')
```

```
plt.subplot(1, 2, 2)
plt.imshow(cv2.cvtColor(watermarked, cv2.COLOR_BGR2RGB))
plt.title('Watermarked Image')
plt.axis('off')

plt.tight_layout()
plt.show()

print("Watermark inserted successfully!")
```

